Task C
CS 4347 - Group 14

create.sql

```
CREATE TABLE CATEGORY (
   CategoryID INT NOT NULL AUTO_INCREMENT,
   CategoryName VARCHAR(100) NOT NULL,
   Description VARCHAR(255) DEFAULT NULL,
   PRIMARY KEY (CategoryID),
   CONSTRAINT uq_category_name UNIQUE (CategoryName)
) ENGINE = InnoDB;

CREATE TABLE PRODUCT (
   ProductID INT NOT NULL AUTO_INCREMENT,
   ProductName VARCHAR(100) NOT NULL,
   PurchasePrice DECIMAL(10,2) NOT NULL,
   SellPrice DECIMAL(10,2) NOT NULL,
   ExpirationDate DATE DEFAULT NULL,
   CategoryID INT NOT NULL,
   PRIMARY KEY (ProductID),
   CONSTRAINT chk_product_purchase CHECK (PurchasePrice >= 0),
   CONSTRAINT chk_product_sell CHECK (SellPrice >= 0),
   CONSTRAINT fk_product_category FOREIGN KEY (CategoryID)
      REFERENCES CATEGORY (CategoryID) ON DELETE RESTRICT
) ENGINE = InnoDB;

CREATE TABLE INVENTORY (
   InventoryID INT NOT NULL AUTO_INCREMENT,
   ProductID INT NOT NULL,
   Quantity INT NOT NULL DEFAULT 0,
   MinStock INT NOT NULL DEFAULT 0,
   LastUpdated TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
      ON UPDATE CURRENT_TIMESTAMP,
   PRIMARY KEY (InventoryID),
   CONSTRAINT uq_inventory_product UNIQUE (ProductID),
   CONSTRAINT chk_inventory_quantity CHECK (Quantity >= 0),
   CONSTRAINT chk_inventory_min_stock CHECK (MinStock >= 0),
   CONSTRAINT fk_inventory_product FOREIGN KEY (ProductID)
      REFERENCES PRODUCT (ProductID) ON DELETE CASCADE
) ENGINE = InnoDB;

CREATE TABLE SALE (
   SaleID INT NOT NULL AUTO_INCREMENT,
   SaleDate TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
   TotalCost DECIMAL(10,2) NOT NULL DEFAULT 0.00,
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    TotalRevenue DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    PRIMARY KEY (SaleID),
    CONSTRAINT chk_sale_cost CHECK (TotalCost >= 0),
    CONSTRAINT chk_sale_revenue CHECK (TotalRevenue >= 0)
) ENGINE = InnoDB;

CREATE TABLE SALEITEM (
    SaleItemID INT NOT NULL AUTO_INCREMENT,
    SaleID INT NOT NULL,
    ProductID INT NOT NULL,
    QuantitySold INT NOT NULL,
    UnitPrice DECIMAL(10,2) NOT NULL,
    UnitCost DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (SaleItemID),
    CONSTRAINT chk_saleitem_quantity CHECK (QuantitySold > 0),
    CONSTRAINT chk_saleitem_price CHECK (UnitPrice >= 0),
    CONSTRAINT chk_saleitem_cost CHECK (UnitCost >= 0),
    CONSTRAINT fk_saleitem_sale FOREIGN KEY (SaleID)
        REFERENCES SALE (SaleID) ON DELETE CASCADE,
    CONSTRAINT fk_saleitem_product FOREIGN KEY (ProductID)
        REFERENCES PRODUCT (ProductID) ON DELETE RESTRICT
) ENGINE = InnoDB;
```

load.sql

```
LOAD DATA LOCAL INFILE 'category.csv'
INTO TABLE CATEGORY
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(CategoryID, CategoryName, @Description)
SET Description = NULLIF(
    TRIM(TRAILING '\r' FROM @Description), '');

LOAD DATA LOCAL INFILE 'product.csv'
INTO TABLE PRODUCT
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(ProductID, ProductName, PurchasePrice, SellPrice,
 @ExpirationDate, CategoryID)
SET ExpirationDate = NULLIF(@ExpirationDate, '');

LOAD DATA LOCAL INFILE 'inventory.csv'
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INTO TABLE INVENTORY
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(InventoryID, ProductID, Quantity, MinStock, LastUpdated);

LOAD DATA LOCAL INFILE 'sale.csv'
INTO TABLE SALE
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(SaleID, SaleDate, TotalCost, TotalRevenue);

LOAD DATA LOCAL INFILE 'saleitem.csv'
INTO TABLE SALEITEM
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(SaleItemID, SaleID, ProductID, QuantitySold, UnitPrice, UnitCost);
```